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A Combinatorial Pigeonhole Manifold Operator for Anastomotic Surfboard Registration

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Abstract

Building upon stochastic Wishart deformations, we present a novel combinatorial topological framework leveraging a continuous analog of the Pigeonhole Principle. By partitioning the target surgical manifold into discrete 'holes' (attractor nodes), source 'pigeons' (excision vertices) are optimally mapped to minimize structural geometric residuals. This reduces the problem to a non-linear Dirichlet energy relaxation gradient, offering deterministically bounded computation time for complex plastic surgery applications and graft modeling.

1. Pigeonhole Combinatorial Map & Dirichlet Energy

For a discrete target manifold Y , we define a subset of attractor 'holes' $\mathcal{H}$. Each source vertex x_i ('pigeon') is assigned to the nearest hole minimizing the L_2 norm:

$$h^*(x_i) = \arg \min_{h \in \mathcal{H}} \|x_i - h\|_2$$

The manifold undergoes a geometric relaxation guided by a discretized Dirichlet energy:

$$E(x) = \frac{1}{2} \sum_i \|x_i - h^*(x_i)\|^2 + \lambda \|x_i - y_i\|^2$$

The iterative update limits asymptotic divergence by linearly blending local combinatorial relaxation with a global target drift vector.

$$x_i^{(t+1)} = x_i^{(t)} + \alpha(h^*(x_i^{(t)}) - x_i^{(t)}) + \beta(y_i - x_i^{(t)})$$

Conclusion

The Pigeonhole Manifold operator provides a robust combinatorial bound to otherwise unconstrained non-rigid deformations, highly optimizing surgical alignments.

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